QWBSACSCORE, and QWBPACSCORE. For example, Johnson *et al.* [31] describe a condition as one day of 'coughing, wheezing, shortness of breath' and 'cannot leave your house, go to work, go to school, do housework, participate in social or recreational activities, and you have some physical limitations (trouble bending, stooping or doing vigorous activities) because of this health condition, but you can care for yourself.' Using the QWB system, this corresponds best with symptom category 11 (QWBSYMSCORE $= 0.257$), mobility scale category 4 (QWBMOBSCORE $=$ 0.062), physical activity scale category 3 (QWBPACSCORE $= 0.060$), and social activity scale category 1 (QWBSACSCORE $= 0.106$) and ΔDAYS $= 1$. Adding the individual QWB scores for this condition, the composite score becomes ΔQWB $= 0.166$. Using this approach, a *higher* score in each of these cases indicates a more severe condition. As shown in Table 3, the symptom scores are on average higher than the other three, but the standard deviation of this score is slightly less. It should be noted that for one study [32], the health effect of interest could not be mapped into the four separate scores, but an overall QWB score was estimated in the study. As a result, one observation is missing for each of the four scores.

The variable ΔQWB provides a summary measure of the reduction in severity of the health effect. It was calculated as the sum of the four scores. This additive assumption is somewhat arbitrary, but it is consistent with the way a total QWB score is typically estimated for a specified health condition. The total QWB score is typically calculated as 1-QWBSYMSCORE–QWBMOBSCORE–QWBSACSCORE–QWBPACSCORE, such that a higher index represents better health. Therefore, ΔQWB is equal to $1-$the total QWB score.

Expanding beyond the Johnson *et al.* approach, we also collected data from each study and included variables to control for other factors that might influence WTP estimates. The variables INCOME, AGE, %MALE, and US were included to account for potentially influential characteristics of the study population. Most studies report summary statistics for these characteristics, but this information is not always reported for the specific subsample that is used to calculate the WTP value. When the subsample information was not available, we used summary statistics for the full study sample.

The dummy variable WTPAVOID was specified to distinguish between WTP values that were estimated for avoiding a decline in health, as opposed to those for improving health from current conditions. A positive effect for this variable would be consistent with declining marginal utility of health. However, a large positive effect for this variable may also indicate deviations from the standard utility model, such as the loss aversion and reference dependence models proposed by Tversky and Kahneman [26].

The remaining dummy variables were included to control for study design effects and for potential publication bias. Finally, we included SAMPLE SIZE, not as an explanatory variable, but as a weighting variable, so that WTP estimates based on larger samples could be given more weight in the regression analysis. The results of including these variables in meta-regressions are described below.

Other factors not specifically included in our list of variables may also influence WTP. For example, prices for medical care, wages (opportunity cost of sick time), and average education are all potentially influential factors. Unfortunately, the amount of information reported in the original studies is not adequate to include these factors in the analysis. However, the income variable may capture part of the wage effect, and the US dummy may serve as a proxy for differences in medical costs between the US and other countries.

Meta-regression models and results

Tables 4 and 5 describe the regression results for several model specifications, all of which share the same basic structure. We included a measure of WTP in all models as the dependent variable. Measures of the change in duration and severity of the corresponding health effect were included as explanatory variables. Additional explanatory variables include characteristics of the study population, valuation method, study design, and publication outlet.

Model specification

To explore the robustness of the results across model specifications, eight sets of results are reported in these tables. Two model specification issues in particular – functional form and health index specification – are addressed in the regres-

Table 4. Weighted least squares meta-regression results with total QWB score ($N = 236$)

Explanatory variable	Dependent variable: WTPACUTE				Dependent variable: ln(WTPACUTE)			
	Coefficient	t-stat[a]	Coefficient	t-stat[a]	Coefficient	t-stat[a]	Coefficient	t-stat[a]
ΔDAYS	4.78	3.59			0.02	4.36		
ln(ΔDAYS)			97.40	4.48			0.50	12.59
ΔQWB	1613.27	2.82			7.23	4.89		
ln(ΔQWB)			454.20	2.23			1.97	3.26
INCOME	0.00	0.57			0.00	1.82		
ln(INCOME)			197.20	1.61			0.70	2.13
AGE	−0.96	−0.05			0.09	2.20		
ln(AGE)			−58.22	−0.08			2.56	1.78
%MALE	−2.90	−0.70	−1.61	−0.58	−0.03	−2.05	−0.01	−1.36
US	18.87	0.14	−41.88	−0.38	−0.19	−0.64	−0.41	−1.48
WTPAVOID	−122.39	−0.34	−43.27	−0.15	0.75	0.99	0.78	1.33
OPEN ENDED	51.77	0.28	93.63	0.84	0.33	0.85	0.20	0.76
PAYMENT CARD	−15.06	−0.16	31.94	0.51	−0.06	−0.22	−0.02	−0.09
IN PERSON	−42.98	−0.33	−124.55	−1.08	−0.11	−0.32	−0.47	−1.28
JOURNAL	43.13	0.26	7.91	0.05	−0.54	−1.37	−0.71	−1.88
CONSTANT	−256.01	−0.22	−1168.54	−0.31	−1.75	−0.64	−10.34	−1.29
R^2(%)	44.47		48.00		57.19		64.48	

[a] Based on robust standard error estimates, corrected for clustering by study.

Table 5. Weighted least squares meta-regression results with four-dimensional QWB scores ($N = 235$)

Explanatory variable	Dependent variable: WTPACUTE				Dependent variable: ln(WTPACUTE)			
	Coefficient	t-stat[a]	Coefficient	t-stat[a]	Coefficient	t-stat[a]	Coefficient	t-stat[a]
ΔDAYS	4.66	3.70			0.02	4.67		
ln(ΔDAYS)			95.09	4.22			0.49	13.15
β[b]			289.81	1.65			1.18	1.61
QWBSYSCORE	−88.88	−0.11	0.50[c]	0.86	1.09	0.39	0.47[c]	0.95
QWBMOBSCORE	2896.72	5.86	2.78[c]	1.34	10.94	9.93	2.73[c]	3.92
QWBSACSCORE	880.54	1.15	0.31[c]	0.93	4.88	1.24	0.45[c]	1.24
QWBPACSCORE	2380.64	1.86			10.49	3.65		
INCOME	0.00	0.62			0.00	2.08		
ln(INCOME)			240.60	1.86			0.88	2.46
AGE	1.93	0.10			0.10	2.70		
ln(AGE)			52.72	0.07			3.04	2.16
%MALE	−5.47	−1.39	−3.87	−1.33	−0.04	−3.84	−0.03	−3.51
US	40.36	0.30	−24.52	−0.21	−0.12	−0.40	−0.35	−1.27
WTPAVOID	−205.62	−0.55	−65.84	−0.22	0.30	0.43	0.57	1.03
OPEN ENDED	40.00	0.22	121.37	1.13	0.33	0.98	0.33	1.48
PAYMENT CARD	10.67	0.12	63.47	0.99	0.08	0.29	0.14	0.68
IN PERSON	−14.55	−0.11	−119.09	−0.91	0.02	0.04	−0.43	−1.07
JOURNAL	44.73	0.29	19.41	0.12	−0.50	−1.31	−0.65	−1.79
CONSTANT	209.85	0.17	−1,978.06	−0.52	0.22	0.09	−13.58	−1.59
R^2(%)	47.07		NA		59.46		NA	

[a] Based on robust standard error estimates, corrected for clustering by study.
[b] See Equation (9) for interpretation of this coefficient.
[c] Refers to η coefficient in Equation (9).

sion results tables. In addition, two model estimation issues – regression weights and clustering effects – were addressed in the meta-regressions and are incorporated in the results tables. Each of these issues is described briefly below. It should also be noted that the robustness of the results described in these tables is further confirmed by the fact that the size and statistical significance of the key variables change very little when the two highest WTP estimates (over $1000 each) are excluded from the regression analysis.

Functional form: To evaluate the robustness of model results, we applied linear, semi-log, and log-linear specifications to analyze the data. Although, all these approaches are reasonable for approximating the relationship between WTP estimates and the other variables described in Table 2, the log-linear approach has a few conceptual advantages. First, it implies that, as changes in severity and duration of illness (and income) approach zero, WTP also approaches zero. Second, it implies that the marginal effect of income on WTP and the marginal effects of the severity and duration changes on WTP are not mutually independent. So, for example, the additional WTP associated with a larger health improvement is not assumed to be independent of an individual's budget. Third, as discussed in more detail below, the log-linear form allows for a more explicit statistical test of the QALY-based valuation approach.

Health index specification. All the results reported in these tables rely on the QWB Scale to describe the severity of acute illness. However, using the composite QWB score by itself implies that the four subcomponent scores – the symptoms, mobility, social activity, and physical activity scores – each have the same marginal effect on WTP. A less restrictive specification allows the four scores to enter the functional relationship separately and independently. In this way it is possible to test directly the simple additive assumption underlying the composite QWB score.

Regression weights: Treating each value estimate equally in the meta-regression is not efficient because it fails to account for the fact that some values are estimated with relatively more precision than others and therefore contribute more information to the analysis. To maximize efficiency, each estimate would ideally be weighted by the inverse of its variance; unfortunately relatively few of the included studies report standard errors or confidence intervals for their WTP estimates. As an approximation of these relative weights, we instead weighted each observation in direct proportion to its sample size (in contrast to the approach used by Johnson *et al*., which used equal weighting).

Panel data modeling: To account for the panel nature of the data, we estimated all the models using clustered robust regression [33]. Because the data used in the analysis are characterized by multiple observations from individual studies, they are likely to violate the ordinary least squares assumptions of independent and identically distributed errors. Therefore, we used the Huber–White method to correct the standard error estimates for potential error correlation within study clusters and unequal variance of errors across clusters [34,35].

Model results

The meta-regression results are summarized in Tables 4 and 5. The two sets of regressions differ primarily with respect to the health index specification.

Results with a composite QWB score. Table 4 reports results for linear, log-linear, and semi-log specifications of WTPACUTE with respect to ΔDAYS and the single composite QWB score (ΔQWB). All the models show a reasonably good fit, with R^2- statistics between 45 and 65%. More importantly, several of the coefficients have the expected sign and are statistically significant.

The results indicate that the WTP estimates pass the scope test. The coefficients for ΔDAYS and ΔQWB, in both linear and log forms, are consistently positive and predominantly significant across specifications at a 0.05 level. In other words, average WTP to avoid acute effects increases with the number of days or acute episodes avoided and with the severity of the conditions avoided. The coefficients for the linear specification indicate that WTP increases by an average of $4.80 for each additional day of acute illness avoided. Therefore, although statistically significant, the estimated marginal WTP for reduced duration of illness is quite small in the linear specification. Unlike the log-linear model, this specification does not restrict WTP to be zero when ΔDAYS is zero, nor does it allow marginal WTP to decline with respect to ΔDAYS. Consequently, marginal WTP may be underestimated for low levels of ΔDAYS. In this model, WTP is estimated to increase by $16 for each 0.01 change (between 0 and 1) in the total QWB score. Using a linear specification implies that these marginal values for ΔDAYS and ΔQWB are constant and independent of one another; therefore, these estimates are best interpreted as average effects across the range of ΔDAYS and

ΔQWB. The log-linear specification discussed below relaxes these restrictions.

As expected, income also has a consistently positive and, for the most part, reasonably significant effect on WTP. Positive income effects support the hypothesis that health is a normal good. The income coefficient in the log-linear specification can be interpreted as an elasticity of WTP with respect to income. The elasticity estimate is 0.7, which implies that income has a positive but relatively inelastic effect on WTP. Notably, controlling for this income effect, the dummy variable for studies done in the United States does not have a significant effect (at a 0.05 level) on WTP in any of these specifications.

The average age of the sample also tends to have a positive effect, although the significance of this variable varies across specifications. To the extent that the age is an inverse proxy for health status, this result is consistent with declining marginal utility of health. In other words, this result suggests that older and less healthy individuals are willing to pay more for an increment in health. The estimated size of this age effect is surprisingly large, however. In the log-linear specification, the elasticity of WTP with respect to age is 2.56.

The coefficient on WTPAVOID is generally not statistically significant. Its sign varies across specifications, although it is more often positive. Therefore, these results do not indicate the presence of loss aversion or strong reference effects.

None of the other variables characterizing the study approach show significant effects on WTP. However, in a few specifications, the JOURNAL coefficient is negative and significant at a 0.10 level, which suggests that values published in peer-reviewed journals are generally lower.

Results with a decomposition of the QWB score. Table 5 reports results for four similar specifications; however, the main difference is that the QWB score is decomposed into its four components.

For the most part, the results using the four QWB scores continue to show significant scope effects. The estimated coefficients for the scores are mostly positive and, particularly for the mobility score, are often statistically significant. For example, the results of the first specification in Table 5 indicate that average annual WTP increases by \$24 and \$29 for each 0.01 increment in the physical activity and mobility scores, respectively.

The decomposition of ΔQWB in these specifications also allows us to test one of the underlying assumptions of the QWB Scale (i.e. that each of the four separate dimensions of the index contributes equally to utility (and thus WTP)). In the linear specification and the semilog specification using ln(WTP) as the dependent variable ΔQWB is linearly decomposed. Separate coefficients are estimated for each of the four scores.

The results of the decomposed model indicate that the mobility score and the physical activity score have larger and consistently significant and positive effects on WTP. The symptom score and, to a lesser extent, the social activity score have lower and generally insignificant effects on WTP. It is worth reemphasizing that the regressions estimate the marginal effects of these scores on *private* WTP. If individuals are directly compensated for lost work time through sick leave policies, then the additional WTP of employers or society to avoid these conditions is not captured by these measures. If, in contrast, the costs of lost work time were to be fully internalized by the individuals experiencing the health effect, then it is likely that the mobility, physical activity, and social activity scores would have a larger effect on private WTP. It is also possible that the relative effects of these scores would be different (e.g. social activity restrictions might have a larger effect on WTP relative to physical activity restrictions).

This finding – that the different scores have different effects on individuals' utility and WTP – contradicts the implicit assumptions of the composite ΔQWB score. F-tests of the restriction that all four of the scores have the same marginal effect on WTP can be rejected at a 0.05 level for each of these specifications.

A comparable analysis of the other specifications is somewhat more complicated. In these cases, a linear decomposition of ΔQWB results in nonlinear models. For example, the simple log-linear model from Equation (7) can be rewritten as

$$\ln(\mathrm{WTP}) = \beta\ \ln(\Delta\mathrm{QWB}) + \beta\ \ln(\Delta\mathrm{DAYS}/365) + \alpha X \tag{8}$$

where X represents the vector of all the other explanatory variables, which act as shifters on the intercept term. When ΔQWB is linearly